Introduction to Algorithms and Data Structures

week 2.2: N-Dimensional Arrays

SAXION UNIVERSITY OF APPLIED SCIENCES

N dimensional Arrays

- **Discuss work last lesson**
- 2D arrays and problem solving
- N-dimensional arrays
- homework

Discussing homework from the last lesson

- Where you able to do the exercises?
  1. Yes! All exercises are done!
  2. No.., Because I have these questions
  3. I haven't prioritized this subject enough at the moment to do the assignments

N dimensional Arrays

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Workshop:

Do exercise: 5_visibleTrees from **n-dimensionalArrays**

You can find this on BrightSpace > week2

N dimensional Arrays

- **Discuss work last lesson**
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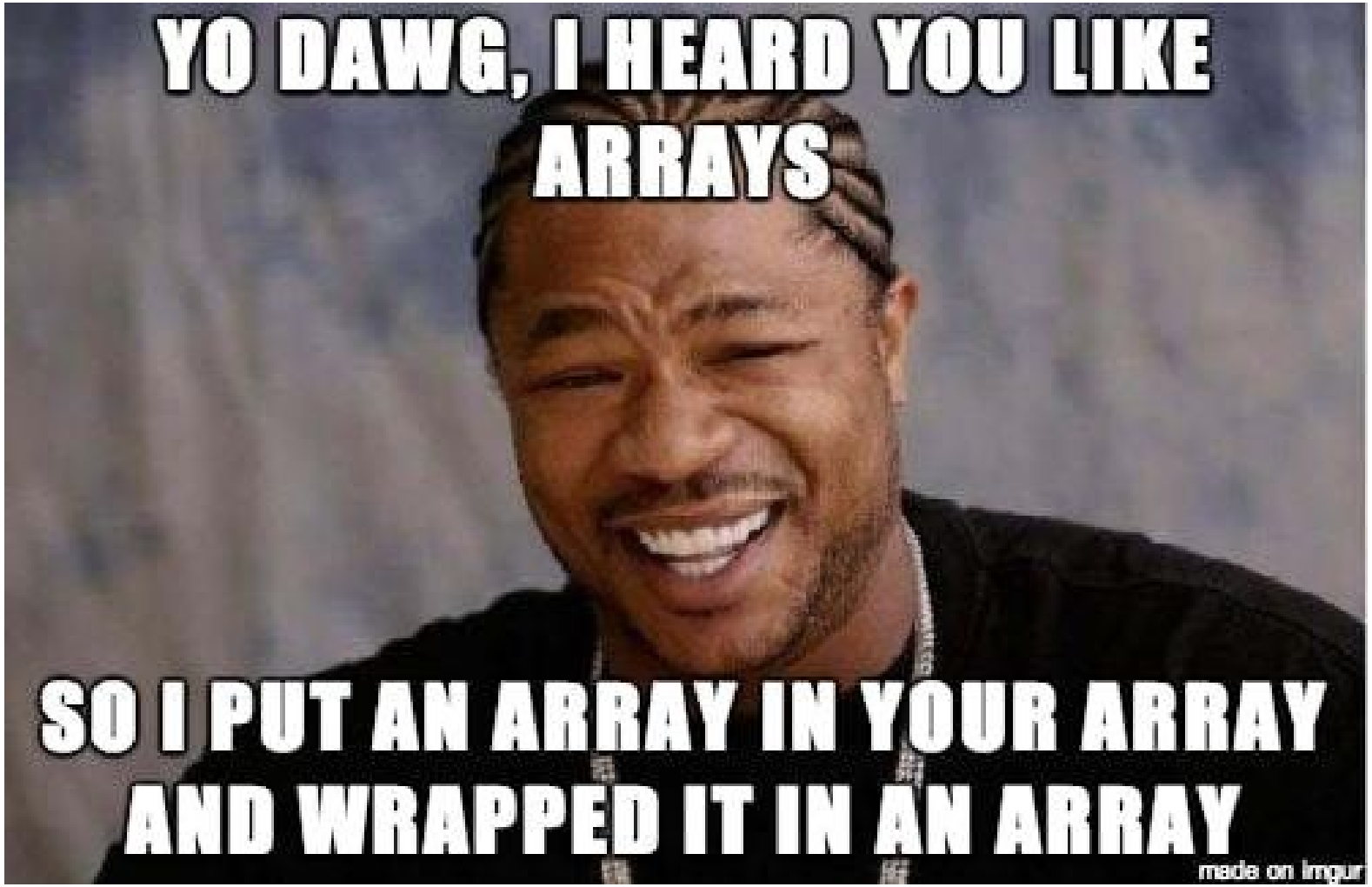
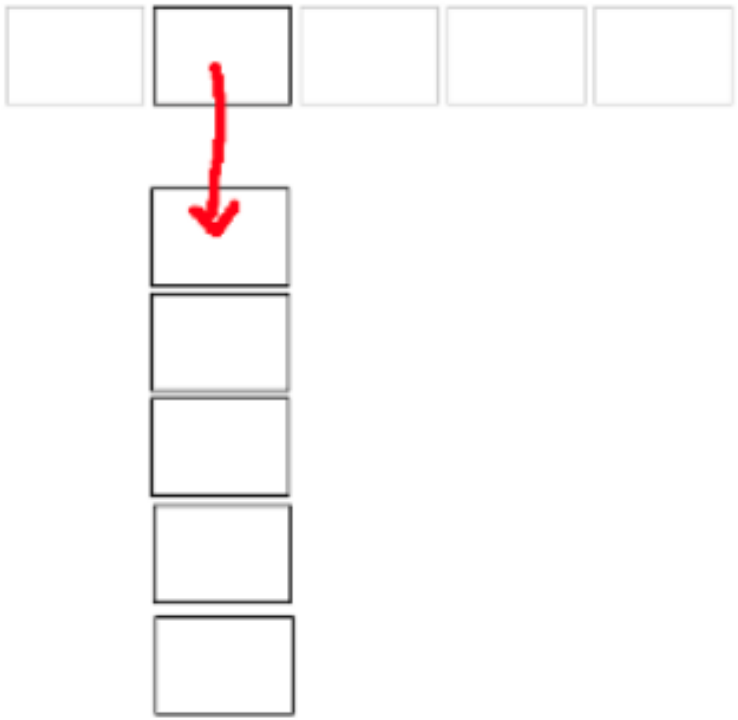
N-dimensional arrays



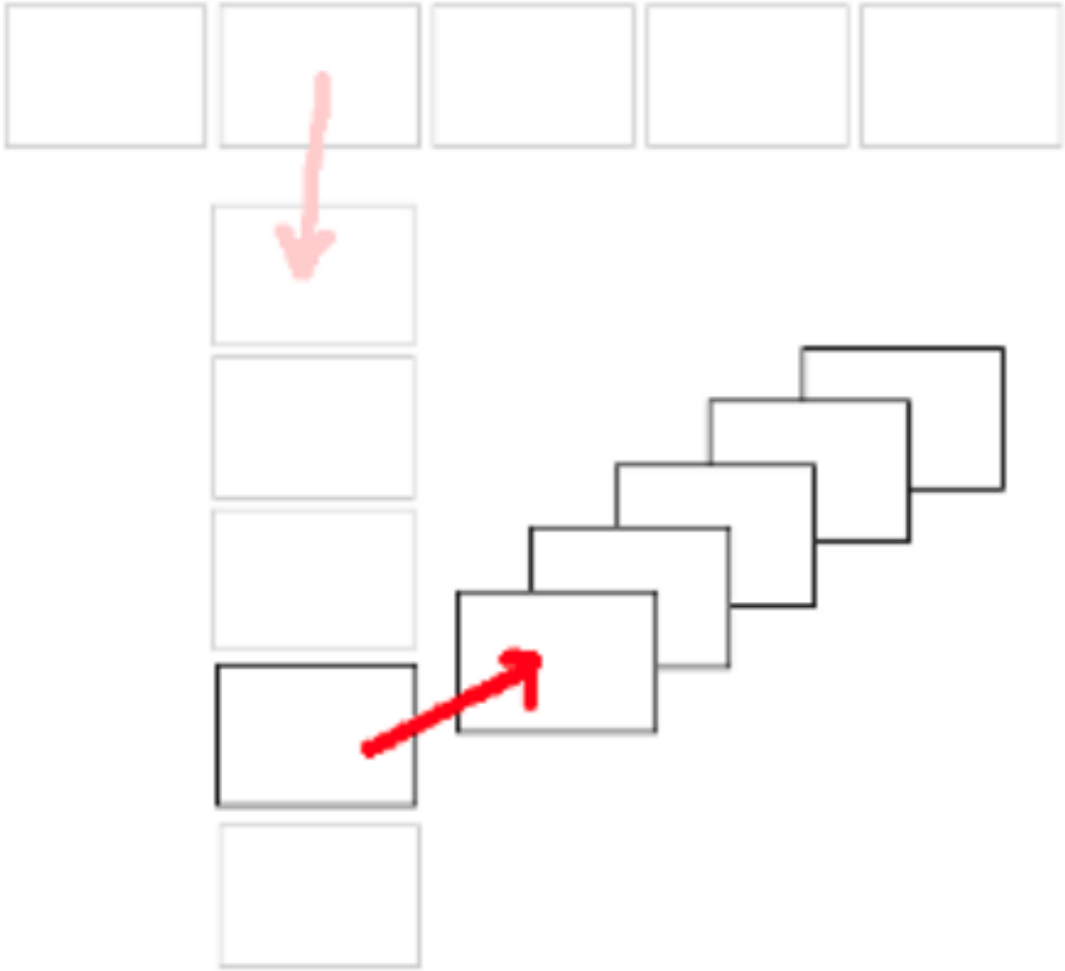
N-dimensional arrays

1-d array

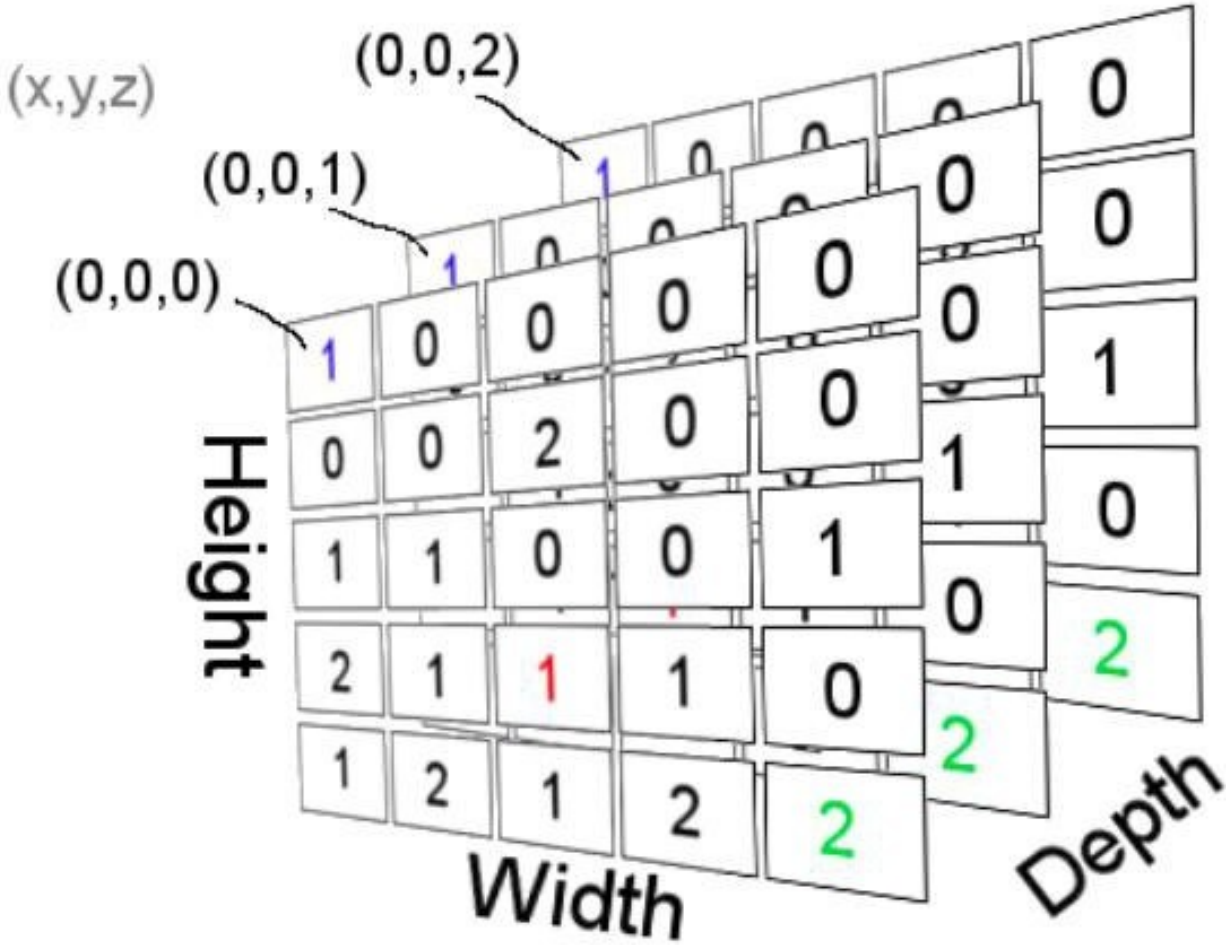
2-d array



3-d array



N-dimensional arrays



N-dimensional arrays

N-dimensional arrays

Day 1

Tijd	Temp	Gevoel	Wind	Buien	Neerslag	Zon
16:00	22°C	22°C	NW 3	1%	0,0 mm	46%
17:00	22°C	22°C	NW 3	0%	0,0 mm	50%
18:00	21°C	21°C	NW 3	0%	0,0 mm	51%
19:00	21°C	21°C	NW 2	0%	0,0 mm	53%
20:00	20°C	20°C	NW 2	0%	0,0 mm	58%
21:00	19°C	19°C	NW 2	0%	0,0 mm	60%
22:00	18°C	18°C	NW 1	0%	0,0 mm	21%
23:00	17°C	17°C	NW 1	0%	0,0 mm	0%

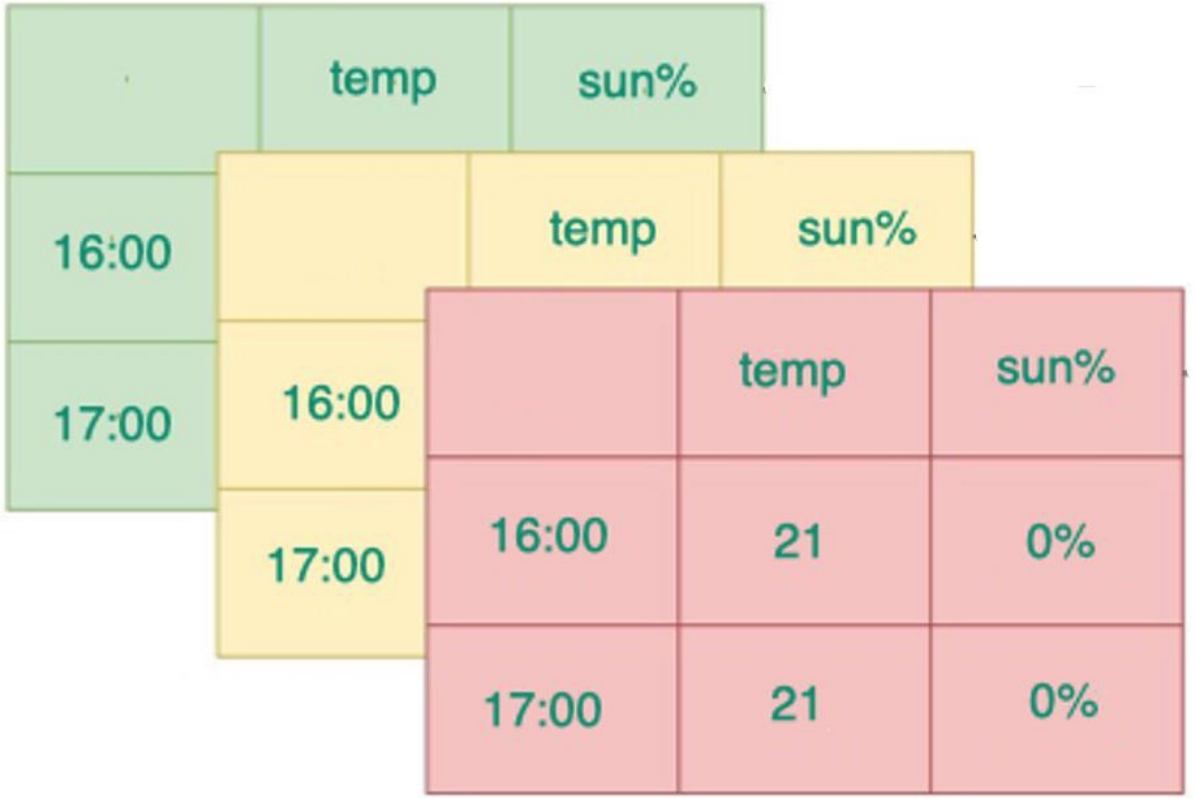
Day 2

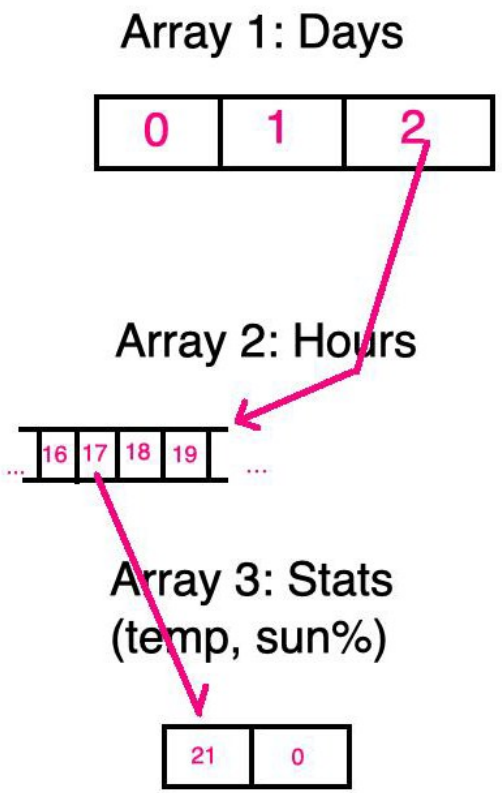
Tijd	Temp	Gevoel	Wind	Buien	Neerslag	Zon
16:00	22°C	25°C	W 0	2%	0,0 mm	6%
17:00	21°C	21°C	W 0	4%	0,0 mm	5%
18:00	21°C	21°C	W 2	1%	0,0 mm	6%
19:00	20°C	20°C	N 1	75%	1,8 mm	11%
20:00	19°C	19°C	N 2	1%	0,0 mm	20%
21:00	18°C	18°C	N 1	0%	0,0 mm	31%
22:00	17°C	17°C	N 1	0%	0,0 mm	38%
23:00	16°C	16°C	N 0	0%	0,0 mm	0%

Day 3

Tijd	Temp	Gevoel	Wind	Buien	Neerslag	Zon
16:00	21°C	21°C	NW 2	0%	0,0 mm	0%
17:00	21°C	21°C	NW 2	0%	0,0 mm	0%
18:00	20°C	20°C	NW 1	0%	0,0 mm	3%
19:00	20°C	20°C	NW 1	0%	0,0 mm	5%
20:00	19°C	19°C	N 1	0%	0,0 mm	11%
21:00	18°C	18°C	NW 0	0%	0,0 mm	15%
22:00	17°C	17°C	NW 0	0%	0,0 mm	6%
23:00	16°C	16°C	N 0	0%	0,0 mm	0%

N-dimensional arrays





Day 1

Tijd	Temp	Gevoel	Wind	Buien	Neerslag	Zon
16:00	22°C	22°C	NW 3	1%	0,0 mm	46%
17:00	22°C	22°C	NW 3	0%	0,0 mm	50%
18:00	21°C	21°C	NW 3	0%	0,0 mm	51%
19:00	21°C	21°C	NW 2	0%	0,0 mm	53%
20:00	20°C	20°C	NW 2	0%	0,0 mm	58%
21:00	19°C	19°C	NW 2	0%	0,0 mm	60%
22:00	18°C	18°C	NW 1	0%	0,0 mm	21%

Day 2

Tijd	Temp	Gevoel	Wind	Buien	Neerslag	Zon
16:00	22°C	25°C	W 0	2%	0,0 mm	6%
17:00	21°C	21°C	W 0	4%	0,0 mm	5%
18:00	21°C	21°C	W 2	1%	0,0 mm	6%
19:00	20°C	20°C	N 1	75%	1,8 mm	11%
20:00	19°C	19°C	N 2	1%	0,0 mm	20%
21:00	18°C	18°C	N 1	0%	0,0 mm	31%

Day 3

Tijd	Temp	Gevoel	Wind	Buien	Neerslag	Zon
16:00	21°C	21°C	NW 2	0%	0,0 mm	0%
17:00	21°C	21°C	NW 2	0%	0,0 mm	0%
18:00	20°C	20°C	NW 1	0%	0,0 mm	3%
19:00	20°C	20°C	NW 1	0%	0,0 mm	5%
20:00	19°C	19°C	N 1	0%	0,0 mm	11%
21:00	18°C	18°C	NW 0	0%	0,0 mm	15%

N-dimensional arrays

- ```
  //create a 3D array
  int[][][] some3DArray = new int[3][3][3];

  //add something
  some3DArray[1][3][2] = 5;

  //get something
  System.out.print(some3DArray[1][3][2]); // 5
  ```
  ```

Workshop:

Do part 1 of exercise: 7_weatherForecast from **n-dimensionalArrays**

You can find this on BrightSpace > week2

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Now it’s your turn!